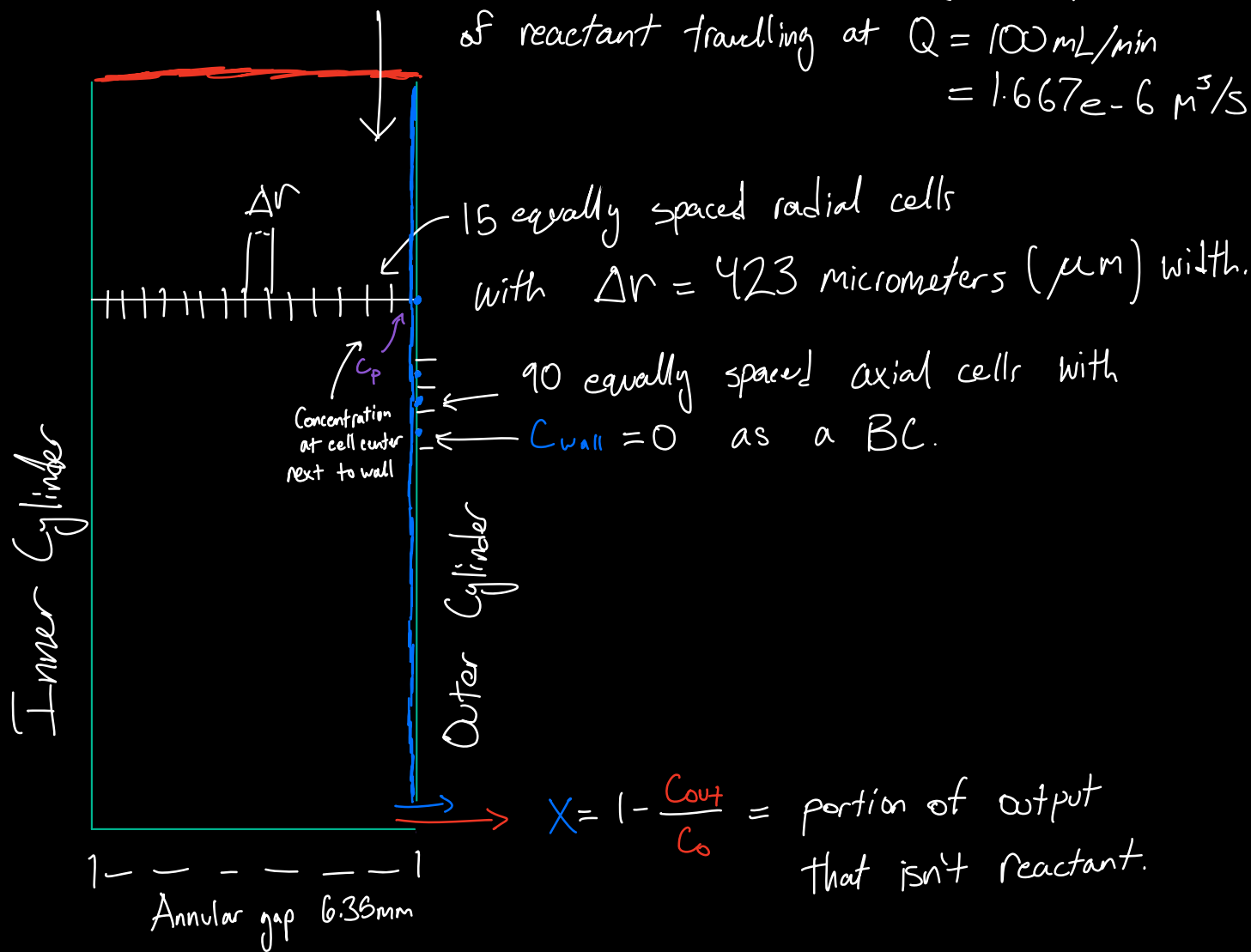


What's going wrong with our radial x axial 15 x 90 side outlet case

Silicone oil mixed with $C_0 = 50$ (mmol/m³) concentration of reactant travelling at $Q = 100$ mL/min
 $= 1.667 \times 10^{-6}$ m³/s



Scalar Transport Equation:

$$\frac{\partial C}{\partial t} + \nabla \cdot (\psi C) - \nabla \cdot (D \nabla C) = 0$$

At steady state $\frac{\partial C}{\partial t} = 0$,

$$\nabla \cdot (\psi C) - \nabla \cdot (D \nabla C) = 0$$

(From here on I'm referencing Incropera Ch.6)

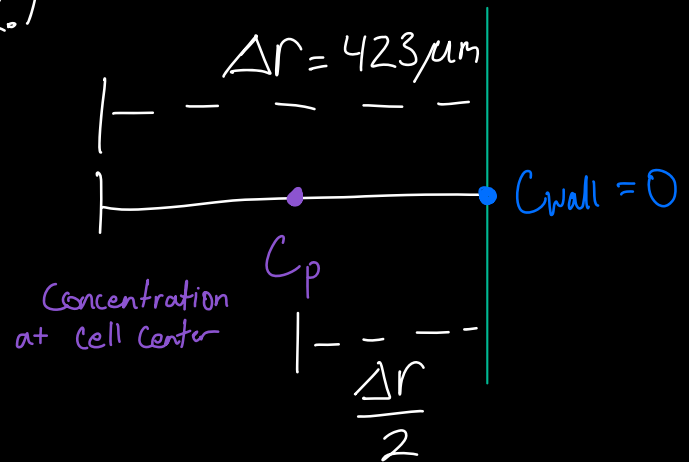
Wall consumption is Fick's Law evaluated at the wall.

$$N_A'' = D \left(\frac{\partial C_A}{\partial r} \Big|_{r=R_0} \right)$$

R_0 is distance from reactor center to outer wall

(N_A'' : Wall Flux of Fluid A per unit area)

OpenFOAM discretizes with



$$N_A'' = D \left(\frac{C_p - C_{wall}}{\Delta r/2} \right)$$

$$= D \left(\frac{C_p - 0}{\Delta r/2} \right)$$

$$= \frac{2DC_p}{\Delta r}$$

If the Concentration Boundary Layer thickness is smaller than $\frac{\Delta r}{2}$ then $C_p \approx C_0$.

$$\text{So } N_A'' = \frac{2DC_0}{\Delta r} \Rightarrow N_A' = \frac{2DC_0}{\Delta r} \cdot A_{wall}$$

How to calculate expected steady-state conversion?

Incropera 6.1.3. mass-transfer coefficient relation.

$$k_c = \frac{D_{AB} \left. \frac{\partial C_A}{\partial r} \right|_{r=R_0}}{C_{\text{wall}} - C_{A,\infty}}$$

I've found a paper by Frank C Walsh for this calculation
Walsh. $X_{sp}^{\text{PFR}} = 1 - \frac{C_{\text{out}}}{C_{\text{in}}} = 1 - e^{-\frac{k_c A}{Q}}$ (Electrode instead of Catalytic wall
However set up is same)
(Equation 12)

For our side outlet geometry we have

$$D = 10^{-9}, \quad \left. \frac{\partial C_A}{\partial r} \right|_{r=R_0} \approx \frac{C_p - C_{\text{wall}}}{\Delta r/2} = \frac{2C_p}{\Delta r},$$

$$\text{Therefore, } k_c = \frac{D \left(\frac{2C_p}{\Delta r} \right)}{-C_{A,\infty}} = \frac{-2DC_p}{\Delta r C_{A,\infty}}$$

If Concentration Boundary Thickness is less than

$$\frac{\Delta r}{2} \text{ then } C_p \rightarrow C_{A,\infty}. \quad \text{Thus, } \underline{k_c^{\text{mesh}} = \frac{-2D}{\Delta r}}.$$

I'm renaming to k_c^{mesh} because this is the k_c

forced by the mesh that is now constant.

Final calculation. $k_c^{\text{mesh}} \Rightarrow X.$

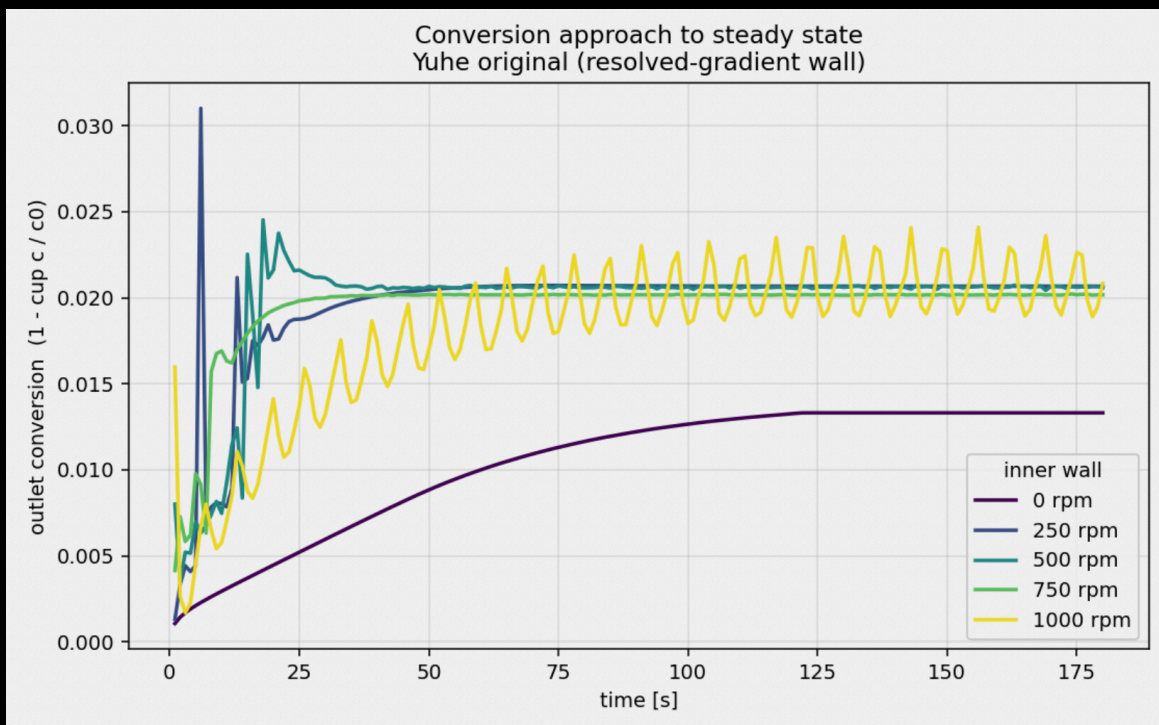
$$D = 10^{-9}, \quad \Delta r = 423.3 \mu\text{m} = 4.233 \times 10^{-4} \text{ m}$$

$$k_c^{\text{mesh}} = \frac{-2(10^{-9})}{4.233 \times 10^{-4}} = -4.725 \times 10^{-6} \text{ m/s}.$$

$$\frac{k_c^{\text{mesh}} A}{Q} = \frac{-(4.725 \times 10^{-6})(7.5 \times 10^{-3})}{1.67 \times 10^{-6}} \quad (A = 2\pi R_o H)$$

$$1 - e^{k_c^{\text{mesh}} A/Q} \approx 0.02099$$

This lines up with our simulation results



~ 0.021

Which is evidence that our issue is under resolution of the concentration boundary layer where $C_p \rightarrow C_{A,\infty}$.

What if Concentration Boundary Layer thickness δ_c is greater than $\frac{\Delta r}{2}$ instead?

Then we get to keep $k_c = \frac{-2D C_p}{\Delta r C_{A,\infty}} < \frac{-2D}{\Delta r}$

because $C_p < C_{A,\infty}$ by assumption.

Key takeaway is that k_c is now difficult to calculate by hand.

Incropera 6.52. $Sh := \frac{k_c d}{D} = \left. \frac{\partial C_A^*}{\partial r^*} \right|_{wall^*}$

AKA Sherwood number is the nondimensionalized gradient.

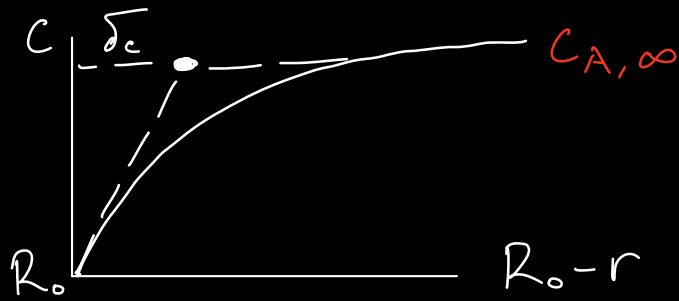
$$k_c = \frac{Sh D}{d}$$

$$\text{With } Sh \approx c Re^a Sc^b, \quad Re = \frac{\omega r_i d}{\nu}, \quad Sc = \frac{\nu}{D}$$

We can get predictions for what conversion

Should be given a angular velocity

Nernst Boundary Layer Thickness.



If we treat the concentration gradient as linear then the Nernst Boundary Layer Thickness is δ_c .

In our OpenFOAM case we do treat the concentration gradient as linear.

So to reach bulk flow concentration $C_{A,\infty}$ it takes boundary layer thickness δ_c meters.

From Bard & Faulkner's *Electrochemical Methods*,
Section 1.4.2. Semiempirical Treatment of
Steady-State Mass Transfer

$$N_A'' = D \left(\frac{\partial C_A}{\partial r} \Big|_{r=R_0} \right)$$

With the assumption of a
linear concentration gradient then

$$N_A'' = \frac{D}{\delta_c} (C_{A,\infty} - C_{wall}).$$

Define $k_c := \frac{D}{\delta_c}$, then

$$N_A'' = k_c (C_{A,\infty} - C_{wall})$$

Therefore in our OpenFOAM simulation

We can estimate δ_c with $\frac{D}{k_c}$
at steady-state.

We can derive two formulas for δ_c .

1) Using Walsh's $X = 1 - e^{\frac{k_c A}{Q}}$

$$\Rightarrow \frac{k_c A}{Q} = \ln(1-X) \Rightarrow k_c = -\frac{Q}{A} \ln(1-X)$$

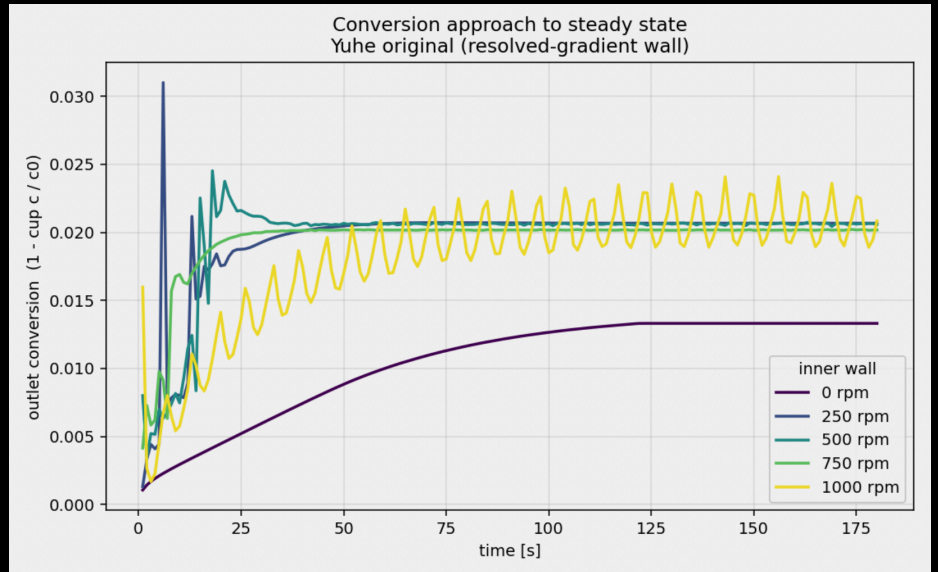
$$\Rightarrow \delta_c = \frac{-DA}{Q \ln(1-X)}$$

2) Using $k_c = \frac{-2D C_p}{\Delta r C_{A,\infty}}$

$$\Rightarrow \delta_c = \frac{\Delta r C_{A,\infty}}{-2 C_p}$$

Formula (1) predictions:

Returning to our simulation.



$$\delta_c = \frac{-(10^{-9})(7.5 \times 10^{-3})}{(1.67 \times 10^{-6}) \ln(1 - 0.0205)}$$

750 RPM got
around $X = 0.0205$

$$= 2.16 \times 10^{-4} \text{ m} \approx 2.11 \times 10^{-4} \text{ m} = \frac{\Delta r}{2}$$

① RPM Run Calculation: $X \approx 0.013$

$$\delta_c = \frac{-(10^{-9})(7.5 \times 10^{-3})}{(1.67 \times 10^{-6}) \ln(1 - 0.013)}$$

$$= -3.43 \times 10^{-4} \text{ m} > 2.11 \times 10^{-4} = \frac{\Delta r}{2}$$

So our boundary layer thickness estimation is consistent with our experimental data.

We are able to confirm that

the 0 RPM run was properly resolved

While all runs > 250 RPM

hit the conversion ceiling

$$X = 1 - e^{\frac{-2DA}{\Delta r Q}}.$$

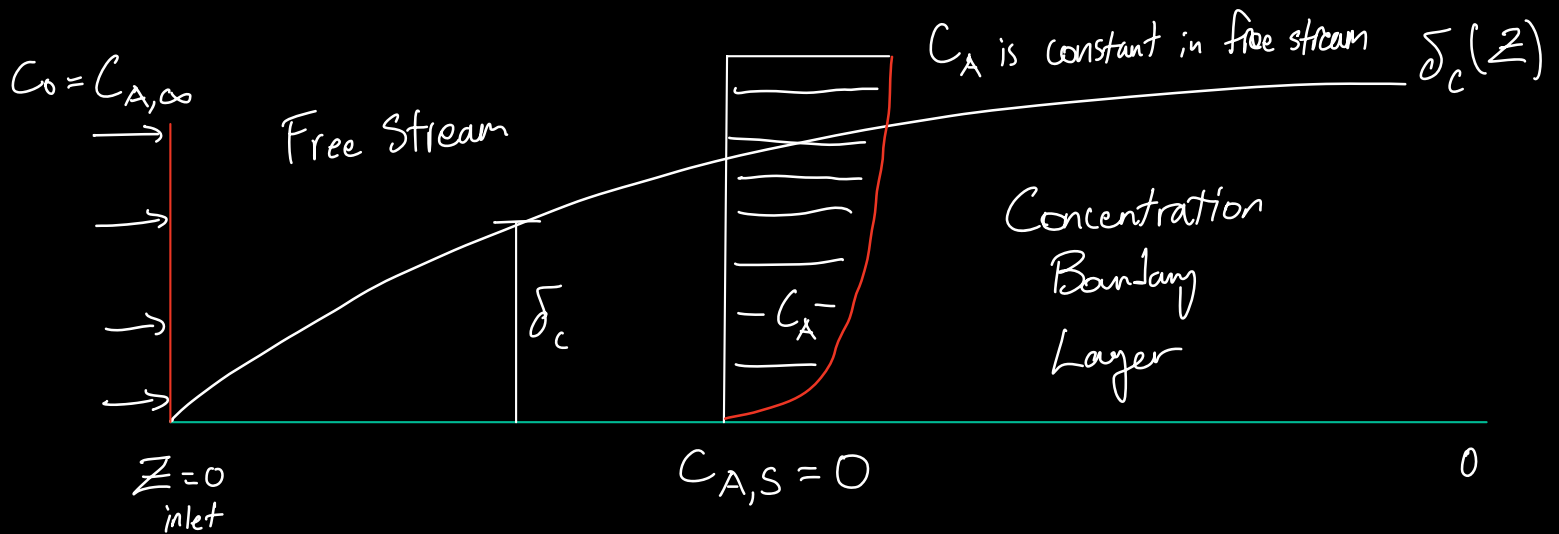
$$\delta_c = \frac{-DA}{Q \ln(1-X)}$$

Now we can estimate steady state conversion

So now the interesting question is how thick is the concentration boundary layer? Is it larger than $\frac{\Delta r}{2}$?

I'm using Incropera's notation.

6.1.3. Concentration Boundary Layer is region of the fluid in which concentration gradient exist.



$\delta_c(z)$ is the point where $C_A = 0.99 C_{A,\infty}$

We will label this as $C_\delta = 0.99 C_{A,\infty} = 0.99 C_0$.

This is me discretizing in the same way as OpenFOAM but with C_δ

$$\left. \frac{\partial C}{\partial r} \right|_{r=R_0} \approx \frac{C_\delta - 0}{\delta_c}$$

Substituting this into Fick's Law gives us

$$N_A'' = -D \left(\frac{c_f}{c_o} \right)$$

I want to show that

$$\text{At steady state, } \int_{\text{wall}} D \left. \frac{\partial c}{\partial r} \right|_{r=R_o} dA = Q c_o X$$